

AETHER HOME – Complete Style Guide (Final Version)

****Project:**** Smart Appliance E-Commerce (React + CSS Modules)

****Based on:**** All provided components (`App`, `Header`, `Footer`, `Home`, `About`, `Products`, `Contact`) and their `.module.css` files.

****Goal:**** This is the definitive, fully corrected style guide covering every visual and interactive detail from the original codebase. No external references needed.

1. Design Tokens (CSS Custom Properties)

Define these inside the root class of every module (e.g., `.home`, `.about`, `.contact`).

- `--accent: #b87c4f` – warm terracotta / gold, used for primary CTAs and active states.
- `--glow: #e2b87a` – light gold, used for hover glows, shadows, and highlights.
- `--metal: #7c8a92` – muted steel blue-grey, used for secondary text and borders.
- `--surface: #2a2a30` – dark grey, used for panel backgrounds and borders.
- `--ink: #f0f0f0` – off-white, the primary text colour.
- `--dark-bg: #1e1e24` – deepest background, base page colour.

Global Background Gradients

- ****Page body**** – `radial-gradient(circle at 20% 30%, #1e1e24, #f0f0f0)`
- ****Hero / CTA sections (darker)**** – `radial-gradient(ellipse at 70% 30%, #1a1a22, #0a0a0e)`
- ****Top bar, Footer, Stats bars**** – `linear-gradient(125deg, #1a1a22, #141418)`
- ****Glassmorphism cards**** – `background: rgba(20,20,26,0.6)` combined with `backdrop-filter: blur(12px)`

Noise Overlay (Home page only)

A fixed overlay with an SVG noise pattern, encoded as a data URI.

```
``css
background-image:
url('
dpZHRoPSIzMDAiIChlaWdodD0iMzAwIj48ZmlsdGVyIGlkPSJmIj48ZmVudXJidWxlbnNlIHR5cGU9ImZyY
WN0YWxOb2l2ZSIgYmFzZUZyZXF1ZW5jeT0iLjciIG51bU9jdGF2ZXM9IjMiLz48L2ZpbHRlcj48cmVjdCB3a
WR0aD0iMTAwIj48Sj48VpZ2h0PSIxMDAlIiBmaWx0ZXI9InVyYyBjZiklIG9wYWNpdHk9IjAuMTUiLz48L3N2
Zz4=')
``
```

2. Typography

Font Families

- ****UI & body text**** – `Inter`, system-ui, sans-serif
- ****Headings & display text**** – `Playfair Display`, serif

Font Sizes (clamped where relevant)

- ****Hero main title**** – Desktop: `clamp(3rem, 9vw, 5.5rem)`; Mobile: `clamp(2.5rem, 8vw, 3.5rem)`
- ****Section title (large)**** – Desktop: `2rem – 2.5rem`; Mobile: `1.6rem – 2rem`
- ****Card name**** – Desktop: `1.3rem – 1.4rem`; Mobile: `1.1rem`
- ****Stat number**** – Desktop: `2.2rem – 2.8rem`; Mobile: `1.8rem – 2rem`

- **Body / description** – Desktop: `0.85rem – 1rem`; Mobile: `0.8rem – 0.9rem`
- **Small meta / badges** – Desktop: `0.6rem – 0.75rem`; Mobile: `0.5rem – 0.65rem`

Gradient Text (Reusable)

```
``css
background: linear-gradient(135deg, #fff, var(--glow), var(--accent));
-webkit-background-clip: text;
background-clip: text;
color: transparent;
``
```

Used on: main titles, logo text, stat numbers, and certain subtitles.

Letter-spacing Animation (for subtitle lines)

```
``css
animation: letterFloat 3s infinite;
@keyframes letterFloat {
  0%, 100% { letter-spacing: 8px; }
  50% { letter-spacing: 14px; }
}
``
```

3. Layout & Spacing

Container Maximum Widths

- Header and product gallery: `1400px`
- Footer: `1300px`
- Timeline and forms: between `900px` and `1200px`

Standard Section Padding

- **Desktop** – Regular sections: `padding: 2rem 8vw`; large sections (timeline, CTA): `padding: 4rem 8vw`
- **Mobile (max-width: 768px)** – Reduced to `padding: 1.5rem 5vw`; tight sections `1rem 5vw`

Responsive Breakpoints

- **Main breakpoint:** `768px` – layout stacks, top bar hidden, grid becomes single column
- **Secondary breakpoint:** `1024px` – minor spacing adjustments, gap reductions

4. Complete Keyframe Animation Library (58 keyframes – all from code)

All keyframes are listed below by category, without any tables.

Entrance & Scroll Reveal

- **fadeUp** – opacity 0→1, translateY(50px), blur(8px) → 0. Duration: 1s. Used on About, Contact, Products hero content.
- **rise** – same as fadeUp but with different easing. Duration: 1s. Used on Contact hero.
- **floatIn** – opacity + translateY(40px). Duration: 0.8s. Used on Home hero text.

- **`titleUnderline`** – underline width 0→100%. Duration: 1.2s. Used on hero titles in About and Contact.

Hover & Attention Effects

- **`iconFloat`** – translateY(-8px) rotate(3deg). 5s infinite alternate.
- **`iconBounce`** – translateY(-5px). 2s infinite.
- **`chipPulse`** – border- color change + translateY(-3px). 4s infinite alternate.
- **`brandFloat`** – translateY(-3px). 2s infinite alternate.
- **`glowSpin`** – rotate(360deg) on `.cardGlow`. 4s infinite linear (on hover only).
- **`pricePulse`** – text- shadow glow. 2s infinite alternate.
- **`pulseOrb`** – scale + box- shadow. 3s infinite alternate. Used on Home featured orb.
- **`particleSpin`** – rotate(360deg) scale(1→1.5). 1.5s infinite. Used on Contact cards.

Continuous Ambient Animations

- **`floatParticle`** – translate, rotate, scale. 30s infinite alternate. Used on Home particles.
- **`driftParticle`** – complex diagonal movement. 25–40s infinite alternate. Used on About, Contact, Products particles.
- **`drift`** – simplified particle movement. 25s infinite alternate. Used on Footer particles.
- **`orbPulse`** – scale(0.9→1.2) + box- shadow. 3s infinite alternate.
- **`spinRing`** – rotate(0→360deg) linear. Duration varies from 8s to 20s per page.
- **`hexSpin`** – rotate(0→12deg) + scale. 6s infinite alternate.
- **`corePulse`** – scale + box- shadow. 2s infinite alternate. Used in Footer and Header logos.
- **`spinCube`** – rotateX/Y 0→360deg. 12s infinite linear. Used on Home rotating cube.
- **`ringOrbit`** – rotate(0→360deg) linear. Duration 10–18s. Used on Contact and Products.
- **`pulseRing`** – scale(0.9→1.2) + fade. 4s ease- out infinite. Used on About.
- **`floatScene`** – translateY + rotate. 6s alternate. Used on Footer 3D scene.
- **`orbFloat`** – translate + scale. 18s infinite alternate. Used on Footer orbs.
- **`hexPulse`** – scale(0.9→1.2) + opacity. 6s infinite alternate. Used on Footer hex.
- **`sparkle`** – letter- spacing animation. 5s infinite. Used on Footer stats.

Component- Specific Unique Keyframes

****Home page****

- ``scrollMarquee`` – infinite horizontal text scroll.
- ``blink`` – cursor opacity blink.
- ``particleFloat`` (cards) – particle moves outward.
- ``audioBounce`` – bar height 5→25px (quote section).
- ``waveAnim`` – bar height 5→35px (animated footer waveform).
- ``rotateWord`` – 3D flip rotation for rotating word.
- ``floatShape`` – translate + rotate for floating hex shapes.

****Products page****

- ``particleBurst`` – diagonal movement + fade.

****Contact page****

- ``sparkFly`` – particle flies from input.
- ``btnParticleFly`` – particle flies from button.
- ``dotPulse`` – dot scales and changes colour.
- ``decorFloat`` – shape moves and rotates (form decoration).
- ``mapCorePulse`` – map core pulse.
- ``mapRingSpin`` – map ring rotation.
- ``mapDotMove`` – map dot movement.

****Header only****

- ``iconPulse`` – ↗ icon pulse.

- `spinSlow` – symbol slow rotation.
- `iconWobble` – contact icons wobble.
- `bgShift` – nav background scale.
- `blinkPulse` – ✂ next to logo.
- `navGlow` – glow under nav item.
- `glowWave` – wave glow on nav hover.
- `beamSway` – beam oscillation.
- `topBarShimmer` – diagonal lines moving.
- `borderPulse` – top bar border pulse.
- `logoSpin` – logo rotation + scale.
- `ringReverse` – reverse ring rotation.
- `pulseExpand` – ripple from click point (used on buttons).

****Footer only****

- `formPulse` – scale pulse on form submit.

5. Pseudo- elements with Text Content

Several pseudo- elements contain decorative text. They are listed below by component.

****Footer****

- `.statsSection::before` – content `🌟 🌟 🌟`, animated with `sparkle` (letter- spacing change).
- `.bottomBar::before` – content `🌟🌟🌟`, purely decorative, no animation.

****Header****

- `.topBar::after` – content `◆◆◆◆◆`, animated with `borderPulse`.
- `.logoText::before` and `.logoText::after` – content `✂`, animated with `blinkPulse`.
- `.contactItem::before` – no content, only a line that animates width on hover.
- `.navItem::before` – no content, underline slide on hover.
- `.navItem .navGlow` – radial gradient that expands on hover.

6. JavaScript- Driven Interactive Effects

All values below are taken directly from the source code.

Tilt Effect (on cards)

- ****Event:**** `onMouseMove`
- ****Formula:****
 - `rotateY = ((mouseX - centerX) / centerX) \* factor`
 - `rotateX = -((mouseY - centerY) / centerY) \* factor`
- ****Factors per component:****
 - Home product cards: 15
 - About value cards: 6
 - About team cards: 8
 - Products cards: 15
 - Contact info cards: 6
- ****Applied style:**** `transform: perspective(1500px) rotateX(...) rotateY(...) scale(1.02)`
- ****Glow:**** The `.cardGlow` radial gradient follows mouse coordinates.
- ****Reset on `onMouseLeave`**** – remove transform and reset glow.

Magnetic Effect (on buttons and social icons)

- **Event:** `onMouseMove``
- **Formula:**
 - `dx = (clientX - rect.left - rect.width/2) * factor``
 - `dy = (clientY - rect.top - rect.height/2) * factor``
 - Clamped to a limit.
- **Values per component:**
 - Header CTA button: factor 0.4, limit ± 20
 - Footer social icons: factor 0.4, limit ± 12
 - Contact form submit button: factor 0.5, limit ± 15
 - Contact social icons: factor 0.5, limit ± 15
 - Home CTA buttons: factor 0.4, limit ± 20
 - Footer CTA button: factor 0.4, limit ± 20
- **Applied style:** `transform: translate(x, y) scale(1.08) rotate(dx * 0.15deg)``
- **Reset on `onMouseLeave`** – remove transform.

Pulse Effect (on click)

- **Event:** `onClick``
- **Calculation:** Get click coordinates relative to button (`clientX - rect.left``).
- **CSS custom properties:** Set `--pulse-x`` and `--pulse-y`` on the button.
- **Add class** `.pulse`` to trigger animation.
- **Animation:** `pulseExpand`` – radial glow expands from click point up to 580px and fades out.
- **Remove class** after 1.2 seconds.

Animated Counter

- **Implementation:** `requestAnimationFrame`` over 2000ms duration.
- **Easing:** Linear.
- **Start:** Immediately on component mount (About, Footer) – no Intersection Observer.
- **Target values:**
 - About page: years 15, devices 50000, clients 48700, awards 38
 - Home page: devices 1840, homes 13200, years 18, awards 67
 - Footer: devices 1840, homes 13200, years 18, awards 67
- **Cleanup:** `cancelAnimationFrame`` in `useEffect``.

Typing Effect

- **Trigger:** `useEffect`` on component mount.
- **Implementation:** `setInterval`` (50–120ms), append one character at a time.
- **Phrases:**
 - About page: "intelligence that adapts to you"
 - Home page: cycles ["AI-powered", "Silent cooling", "Self-cleaning", "Wi-Fi 6 ready"]
- **Cursor:** Uses `blink`` animation on a separate span.

Countdown Timer (Products page)

- **Trigger:** `setInterval`` every 1 second.
- **Target date:** Current date + 3 days, at 23:59:59.999.
- **Output:** Days, hours, minutes, seconds.
- **Stop when zero.**

Mouse Trail (Home page only)

- **Event:** `window.addEventListener("mousemove")``
- **Storage:** Array of up to 10 points `{ x, y, id }``.
- **Render:** Small circular dots with `opacity: 0.8 - i * 0.08``.
- **Cleanup:** automatic as array size limit.

Global Mouse Glow (all pages)

- **Event:** ``mousemove`` on ``window``.
- **Storage:** ``mousePos { x, y }`` state.
- **Application:** Two fixed `<div>` elements with large blur, ``mix-blend-mode: overlay``, their ``left`` and ``top`` follow ``mousePos``.
 - First glow: size 400px, background ``var(--glow)``, blur 60px, opacity 0.3
 - Second glow: size 250px, background ``var(--accent)``, blur 40px, opacity 0.2

7. Scroll Reveal Mechanisms (per page)

Home Page

- **Method:** ``window.addEventListener("scroll")`` + ``getBoundingClientRect()``.
- **Condition:** element's `top < window.innerHeight * 0.75``.
- **Class added:** ``.revealed`` (CSS transition on opacity + transform).
- **Elements affected:** signature, featured, testimonials, stats, quote.

About Page

- **Method:** Intersection Observer with ``threshold: 0.2``.
- **Class added:** ``.visible``.
- **Elements affected:** timeline items, team cards, value cards.

Products Page

- **Method:** Intersection Observer with ``threshold: 0.1`` and ``rootMargin: "50px"``.
- **State:** ``visibleCards`` array stores indices of visible cards.
- **Class added:** ``.visible`` (CSS transition on opacity + transform).
- **Additionally:** ``transition-delay: idx * 0.05s`` on each card.

Contact Page

- **Method:** Intersection Observer with ``threshold: 0.2``.
- **Class added:** ``.visible``.
- **Elements affected:** info cards.
- **Additionally:** ``transition-delay`` values ``0s, 0.3s, 0.6s, 0.9s`` respectively.

8. Particle Field Details

All particle fields are described below without tables.

- **About page** – 80 particles, generated with ``Math.random()``, animated with ``driftParticle``, border-radius ``2px 8px 4px 12px``, blur 3px, opacity 0.3.
- **Contact page** – 80 particles, ``Math.random()``, ``driftParticle``, same border-radius, blur 3px, opacity between 0.1 and 0.3.
- **Products page** – 80 particles, generated with ``createSeededRandom(12345)`` for reproducibility, ``driftParticle``, same border-radius, blur 3px, opacity 0.25.
- **Footer** – 100 particles, ``Math.random()``, animated with ``drift``, border-radius ``1px 4px 2px 6px``, blur 2px, opacity between 0.2 and 0.5.
- **Home (footer 3D scene)** – 30 particles, ``Math.random()``, animated with ``floatParticle``, border-radius 50% (circles), blur 1px, opacity between 0.2 and 0.7.

Note: Seeded random in Products ensures particles are identical between renders.

9. Staggered Delays (Pseudo- random / incremental)

All delays are implemented via inline styles or the `style` attribute.

- **Home hero title letters** – `animation-delay: i \* 0.08s` (each letter individually).
- **Products cards** – `transition-delay: idx \* 0.05s`.
- **Contact info cards** – `transition-delay` values `0s, 0.3s, 0.6s, 0.9s`.
- **Home audio wave bars** – `animation-delay: i \* 0.1s`.
- **Home animated footer waveform** – `animation-delay: i \* 0.05s`.
- **Home floating shapes** – `animation-delay: i \* 0.6s`.
- **Brand strip icons** – `animation-delay: i \* 0.2s`.
- **Footer particles** – `animation-delay: Math.random() \* 20s`.

10. Responsive Details (beyond basic breakpoints)

At max-width: 768px

Global changes

- Hide `.topBar` in Header.
- Reduce section padding to `1.5rem 5vw`.
- Many containers become `flex-direction: column`.
- Card grid becomes `grid-template-columns: 1fr`.
- Font sizes reduce as described in typography.

Home page specific

- `.geometricShapes` { transform: scale(0.7); }
- `.waveform` { transform: scale(0.8); }
- `.audioWave` { transform: scale(0.8); }
- `.heroCanvas` { height: 240px; }

Products page specific

- `.countdownBlock` { min-width: 50px; padding: 0.3rem 0.6rem; }
- `.countdownBlock span` { font-size: 1.2rem; }
- `.cardGrid.listMode .card` { flex-direction: column; text-align: center; }
- `.cardGrid.listMode .card .cardInner` { flex-direction: column; }

Contact page specific

- `.mapPlaceholder` { padding: 1.5rem; min-height: 280px; }
- `.heroCanvas` { height: 260px; }
- `.infoCards, .formSection, .mapSection, .social` { padding: 2rem 5vw; }

Footer specific

- `.statsSection` { flex-direction: column; align-items: center; gap: 15px; }
- `.inputGroup` { flex-direction: column; padding: 12px; border-radius: 1.5rem; }
- `.subscribeBtn` { width: 100%; justify-content: center; margin-top: 8px; }
- `.bottomBar` { flex-direction: column; text-align: center; }
- `.socialIcons` { justify-content: center; }
- `.copyright` { text-align: center; }

At max-width: 1024px

- Header navigation gap: from 45px to 30px.
- Logo text: from 2rem to 1.8rem.
- Contact group gap: from 30px to 20px.
- Adjust padding to `2rem 6vw` on some sections.

11. Reduced Motion Support (`prefers-reduced-motion`)

****Current status in code:**** ✗ Not fully implemented in CSS.

- `App.jsx` has a `reducedMotion` state and a toggle button that adds class `.reduced-motion` to the app container.
- ****No CSS rules exist**** for `.reduced-motion` or `@media (prefers-reduced-motion: reduce)` in any `.module.css` file. Therefore animations never get disabled.

****Required implementation (to be added by developer):****

```
```css
.reduced-motion *,
.reduced-motion *::before,
.reduced-motion *::after {
 animation-duration: 0.01ms !important;
 animation-iteration-count: 1 !important;
 transition-duration: 0.01ms !important;
 scroll-behavior: auto !important;
}
```
```

Alternatively, wrap each animation in a media query:

```
```css
@media (prefers-reduced-motion: no-preference) {
 /* all animations go here */
}
```
```

12. List Mode for Products (Detailed)

When `.listMode` is added to `.cardGrid`:

- `.card` becomes `display: flex; flex-direction: row; align-items: center; gap: 2rem; padding: 1rem;`
- `.cardInner` becomes `flex: 1; display: flex; align-items: center; flex-wrap: wrap; gap: 1rem;`
- `.cardIcon` loses its bottom margin (`margin-bottom: 0`) and sits inline with the text.
- On mobile (`max-width: 768px`), the card returns to column layout.

13. Implementation Checklist for Developers

- Use ****CSS Modules**** – each component imports the `styles` object from its own `.module.css` file.
- Define the design tokens at the top of every module, inside the root class (e.g., `.home`).
- For tilt and magnetic effects, use `useRef` to reference DOM elements and attach `onMouseMove` / `onMouseLeave` handlers.

- For counters and timers, always cancel `requestAnimationFrame` or `clearInterval` inside the `useEffect` cleanup function to prevent memory leaks.
- Prefer `transform` over `top/left` for better GPU acceleration.
- Use `clip-path` to create hexagons and polygons (e.g., for logos and decorative shapes).
- Use `mix-blend-mode: overlay` for glow effects to create subtle lighting.
- Generate particles with `Math.random()`, unless reproducibility is required (then use a seeded random function like `createSeededRandom`).
- Ensure `onMouseMove` handlers are throttled if performance becomes an issue.
- Add the missing `prefers-reduced-motion` CSS as described in section 11.
